

ECO 362 Financial Investments

Midterm Exam

Prof. Motohiro Yogo

Fall 2023

Instructions

1. Check that you have all 4 pages. There are 4 questions.

Question	Points
1	25
2	25
3	20
4	30
Total	100

2. This is a closed book exam. However, you may use a one-page cheat sheet containing formulas or anything else you may find helpful. The cheat sheet must be letter size (8.5x11in) with writing on one side only (i.e., the back side must be left blank). You are under the Honor Code to comply with this requirement.
3. You may use a writing instrument, a cheat sheet, and a calculator. Put everything else away. The calculator cannot be a phone, a tablet, or a laptop. If you do not have a calculator, you may borrow one. Please return it after the exam.
4. You may not communicate with other students or violate the Honor Code during this exam. Write your name and sign the Honor Code on the first page of the exam book.
I pledge my honor that I have not violated the Honor Code during this examination.
5. Write your answers in the exam books. *Show all steps necessary in deriving your solution for full credit.* Report the yield, the coupon rate, or the return in percentage points. Round all answers to two decimal places (e.g., \$95.67 or 1.23%).
6. You have 90 minutes to complete the exam. After 90 minutes, you must stop working on the exam. You then have an extra 20 minutes to submit your answers as one scanned pdf file or photographed images through Gradescope. If you do not have a device to scan or photograph, give your exam books to a preceptor, who will be able to do it for you.
7. In addition to submission by Gradescope, submit the hard copies of the exam books before you leave the room. We need them in case there are any technical issues (e.g., unreadable scans). Make sure that your name is on the exam books that you submit.

1. The following table reports continuously compounded forward yields in percentage points in 2021. For example, the continuously compounded yield on a loan from year 2 to 3 (i.e., 2023 to 2024) is 1.44%. Assume no arbitrage.

Period	Yield (%)
(0,1)	0.42
(1,2)	1.10
(2,3)	1.44
(3,4)	1.66
(4,5)	1.75

[25 points total: 5 points each part]

- (a) In 2021, what is the 2-year zero coupon yield?
 - (b) In 2021, what is the 3-year par yield?
 - (c) In 2021, what is the price of a 2-year annuity that pays \$100 in 2022 and 2023?
 - (d) In 2021, what is the continuously compounded yield on a loan from year 2 to 5 (i.e., 2023 to 2026) ?
 - (e) If you make a riskless investment of \$10,000 in 2021, what is your wealth in 2026?
2. There are two possible states a year from now: expansion or recession. The following table contains the current price and the possible future prices of four stocks a year from now. Assume no arbitrage.

[25 points total: 5 points each part]

Stock	Current price	Possible future prices	
		Expansion	Recession
Amazon	\$126	\$164	\$113
Apple	\$172		\$138
Nvidia	\$442	\$619	\$354
Tesla	\$247	\$210	

- (a) What is the price of a security that pays \$1 in expansion and \$0 in recession.
- (b) What is the price of a security that pays \$0 in expansion and \$1 in recession.
- (c) What is the 1-year annually compounded riskless interest rate?
- (d) What is the future price of Apple in expansion?
- (e) What is the future price of Tesla in recession?

3. The following table contains the annually compounded returns on the riskless asset and stocks in percentage points. Stock returns have a mean of 12% and a standard deviation of 20%.

Year	Riskless (%)	Stocks (%)
2006	4.80	15.40
2007	4.66	5.70
2008	1.60	-36.74

[20 points total: 5 points each part]

- (a) You invest \$1,000 in only stocks at the beginning of 2006, and you are always 100% in stocks. What is your wealth at the end of 2008 (after three years of returns)?
- (b) For what value of risk aversion γ is the optimal portfolio 60% in stocks and 40% in the riskless asset at the beginning of 2006?
- (c) You invest \$1,000 split into 60% in stocks and 40% in the riskless asset at the beginning of 2006, and you rebalance to the 60/40 portfolio at the end of each year. What is your wealth at the end of 2008 (after three years of returns)?
- (d) Based on your answers to parts (a) and (c), explain why a portfolio that is only in stocks may not be optimal for some investors.

4. The following table reports the mean and standard deviation of returns for three assets. The riskless asset has a constant return. The correlation between USA and Canada is zero.

Asset	Mean (%)	Standard deviation (%)
Riskless	0.08	0
USA	0.87	4.30
Canada	0.89	5.65

Assume that the investor has mean-variance expected utility:

$$\mathbb{E}[R_p] - \frac{\gamma}{2} \text{Var}(R_p),$$

where R_p is the portfolio return.

[30 points total: 6 points each part]

- Compute the optimal portfolio for an investor with risk aversion $\gamma = 10$. Report the portfolio weights for all three assets.
- What is the Sharpe ratio for the optimal portfolio from part (a)? Recall that the Sharpe ratio is the ratio of mean excess returns to the standard deviation of returns:

$$\frac{\mathbb{E}[R_p - R_b]}{\sigma(R_p)},$$

where R_p is the portfolio return and R_b is the riskless interest rate.

- If the correlation between USA and Canada were 0.8 (instead of zero), would the Sharpe ratio be higher or lower than your answer from part (b)? To receive credit, explain the reasoning for your answer. You may use a figure to illustrate your reasoning, but no math is necessary.
- If the riskless rate were 1% (instead of 0.08%), would the Sharpe ratio be higher or lower than your answer from part (b)? To receive credit, explain the reasoning for your answer. You may use a figure to illustrate your reasoning, but no math is necessary.
- Compute the optimal portfolio for an investor with risk aversion $\gamma = 15$, using your answer from part (a) and the portfolio separation theorem. Report the portfolio weights for all three assets. You will not get any credit for this question if you do not apply the portfolio separation theorem.